

Raw RDB

User Table

User ID	Account Level	Purchase Amount
1	Gold	100
2	Silver	80
3	Bronze	10

Purchase Table

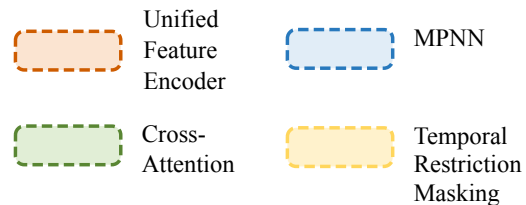
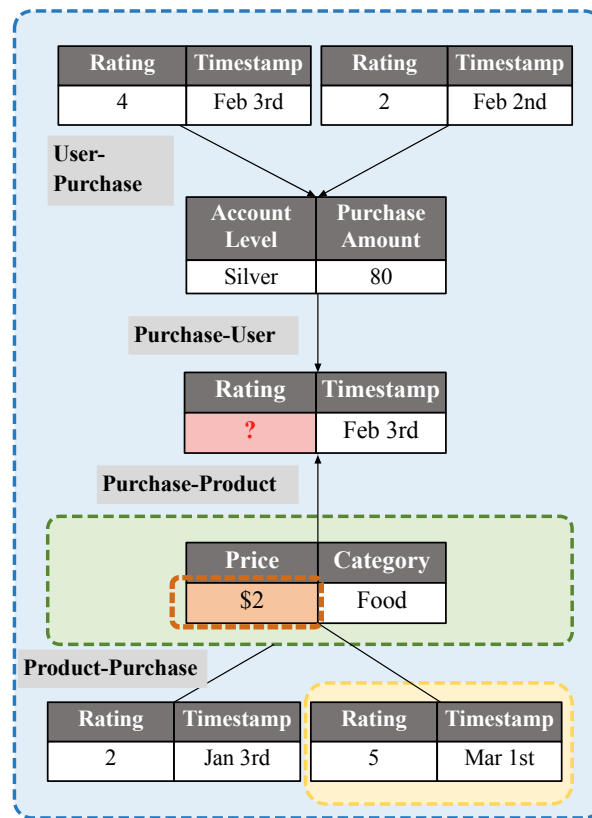
User ID	Item ID	Rating	Timestamp
1	2	5	Jan 1st
1	3	3	Jan 2nd
1	4	2	Jan 3rd
2	1	4	Feb 1st
2	2	2	Feb 2nd
2	4	?	Feb 3rd
3	4	5	Mar 1st

Product Table

Item ID	Price	Category
1	\$10	Book
2	\$5	Food
3	\$30	Furniture
4	\$2	Food

Task: What's the rating of a given purchase record?

Sample Subgraph



Model Architecture

